Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 20)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
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6	
7	
8	
9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Which of the following habitats has low biodiversity but a high proportion of endemic species? Tick one box.

[1 mark]

A Tropical rainforest ☐

B Oceanic islands ☐

C Temperate broadleaf woodland ☐

D Mangroves ☐

0 1 . 2 Compare the ecological features of tropical rainforest and Antarctica that explain their very different biodiversity.

[4 marks]

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0 2 . 1 Explain why a moderate water shortage reduces crop productivity even though the plants survive.

[3 marks]

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0 2 . 2 Explain the problems caused by waterlogged soils and describe **two** ways farmers reduce soil water levels.

[4 marks]

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0 2 . 3 Explain how irrigating with groundwater can cause soil salinisation and how this harms crops.

[2 marks]

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0 2 . 4 State how a farmer could raise the pH of an acidic soil and explain why a low pH reduces fertility.

[1 mark]

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0 3 . 1 Explain the difference between primary and secondary succession and why secondary succession is faster.

[3 marks]

0 3 . 2 Explain what is meant by **deflected succession** and a **plagioclimax**, using a named UK example.

[3 marks]

0 3 . 3 Explain why conservation of a plagioclimax requires continued human activity whereas conservation of a rainforest or coral reef does not.

[2 marks]

0 3 . 4 Explain why biodiversity tends to be low in abiotically extreme environments and high in stable ones.

[2 marks]

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0 4 Table 1 lists some techniques used in ecological monitoring.

0 4 . 1 Complete Table 1 by giving one application and one limitation of each technique.

[6 marks]

Table 1

Technique	Application	Limitation	
Image databases of unique markings			
eDNA sampling			
Geolocator tags			

0 4 . 2 Explain how satellite radar (microwave) sensors are useful for monitoring oil pollution and sea conditions.

[2 marks]

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0 4 . 3 Give **two** types of indirect evidence of an animal's presence and the information each can give.

[2 marks]

1 .....

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0 5 A new lowland nature reserve is being created on former farmland next to an existing ancient woodland.

0 5 . 1 Explain why the **area** and the **shape** of the reserve both matter in its design.

[3 marks]

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0 5 . 2 Explain the value of a biological corridor linking the new reserve to the ancient woodland.

[2 marks]

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0 5 . 3 Explain how each of the following abiotic factors could be managed to increase the biodiversity of the reserve.

[4 marks]

Water levels .....

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Soil nutrient levels .....

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Topography .....

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Light levels .....

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0 5 . 4 Much of the reserve will be kept as a **plagioclimax**. Explain what this means and how grazing is used to maintain one.

[3 marks]

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0 5 . 5 Explain why the removal or culling of some species may be necessary on the reserve.

[3 marks]

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0 6 . 1 Describe **three** methods of reducing seabird by-catch in long-line fisheries.

[3 marks]

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0 6 . 2 Explain how restricting fishing effort and restricting fishing methods reduce overfishing.

[3 marks]

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0 6 . 3 Describe how captive rearing and release ('population seeding') can increase wild populations of lobsters or oysters.

[2 marks]

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0 6 . 4 Explain why collecting data on planktonic fish eggs and larvae helps to prevent population collapse.

[2 marks]

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0 7 . 1 Explain the meaning of each of the following terms and explain why such species are given conservation priority.

[8 marks]

EDGE species

Endemic species

Keystone species

Flagship species

0 7 . 2 Explain why the degree of population dispersal is considered when setting priorities.

[2 marks]

10

0 8 Table 1 shows the ecological footprint and biocapacity per person for four countries and the world in 2016.

Table 1

	Ecological footprint / gha per person	Biocapacity / gha per person	Human Development Index
USA	8.1	3.6	0.92
UK	4.4	1.4	0.92
China	3.6	1.0	0.75
Bangladesh	0.8	0.4	0.61
World	2.8	1.7	0.72

0 8 . 1 Define **ecological footprint** and **biocapacity**.

[2 marks]

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0 8 . 2 Explain what is meant by a **global hectare** and why it is used.

[2 marks]

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0 8 . 3 Use Table 1 to calculate how many Earths would be needed if everyone lived like the average US citizen.

[1 mark]

Answer _____ Earths

Do not write outside the box

0 8 . 4 Use Table 1 to describe the relationship between development and ecological footprint, and explain the concept of Earth Overshoot Day.

[3 marks]

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0 8 . 5 Suggest **two** ways a country such as the UK could reduce its footprint without reducing its HDI.

[2 marks]

1 .....

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0 9 . 1 Which of the following would be most suitable for monitoring the movements of a whale over several months? Tick one box.

[1 mark]

A Radio collar ☐

B Acoustic / satellite tag ☐

C Pitfall trap ☐

D Geolocator tag recovered by recapture ☐

0 9 . 2 Explain why post-release monitoring is important in a captive breeding and release programme and describe **two** suitable methods.

[4 marks]

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1 0 . 1 Discuss the strategies that can be used to increase the sustainability of agriculture.

[9 marks]

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Extra space

1 0 . 2 State what is meant by **permaculture**.

[1 mark]

10

Write an essay on one of the following topics.

11 . 1 Discuss the extent to which the resources and services provided by other species justify the conservation of biodiversity.

[25 marks]

OR

11 . 2 Using tropical coral reefs as an example, discuss how the ecological features of a habitat determine both its importance and its vulnerability to human activities.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS